MC 451 Data Wrangling in R rubric

50 points. Due Friday, November 13, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 50. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Submissions are unlimited before the deadline, so you can revise and submit again.

- Inferencing Data needs all four kinds of variable from this table: two categorical, one with three or more groups, and two continuous. Step 5 is where they are kept.
- What it feeds: Describing Data and Inferencing Data both read `twitch_analysis.RDS`. Your White Paper reuses the moves in step 1 (read a coding form in, make the codes factors, join by id) on your own data; the Twitch table itself stays here.
- Before you submit: restart R, run the script top to bottom, and see what breaks.

Data Wrangling in R, 50 points

Criterion	Exemplary	Proficient	Developing	Beginning
1. Hand-code, and read your codes back 6 points	**6 points.** One variable from your codebook hand-coded for a 30-message sample drawn with `sample_messages()`, recorded in a coding-form spreadsheet (one row per message, one column per variable), read back with `read_csv()`, its column names tidied with `clean_names()`, the codes turned into a labeled factor with `factor()`, and joined to the sample by message id.	**5 points.** Hand-coded and read back, but the codes stay as bare numbers or text with no `factor()`, or the sample was not drawn with `sample_messages()` or is not 30 messages.	**3 to 4 points.** Coded with a variable that is not in your codebook, or the codes are typed into the script instead of read from a coding form.	**0 to 2 points.** No hand-coding.
2. Load 6 points	**6 points.** The chat and stream tables are loaded from the v2v package inside the script.	**5 points.** Both loaded, but from a file on your own computer, so the script fails anywhere else.	**3 to 4 points.** Only one table loaded.	**0 to 2 points.** Not loaded in the script.
3. Convert the timestamps 6 points	**6 points.** Epoch milliseconds converted to a real date-time, dividing by 1,000 first.	**5 points.** Converted, but a small error remains, such as times off by some hours because the time zone was not set.	**3 to 4 points.** Converted without dividing by 1,000, so the dates land thousands of years away.	**0 to 2 points.** Not converted.
4. Message length 6 points	**6 points.** A new column holding each message's length.	**5 points.** Length computed, but on the wrong field (for example, the username).	**3 to 4 points.** Length computed but not saved as a column.	**0 to 2 points.** Missing.

Criterion	Exemplary	Proficient	Developing	Beginning
5. Join, and keep the analysis columns 6 points	**6 points.** Stream information joined onto the chat messages, with a comment naming the key you joined on, and the table keeps the channel's main category (game), `is_gaming`, `viewer_band`, and one continuous audience measure (`median_viewers`, or each message's viewers from the snapshot nearest before it).	**5 points.** Joined and the columns kept, but no comment names the key, or the continuous audience measure is missing.	**3 to 4 points.** The join runs but matches nothing or multiplies the rows, and the script does not notice; or only `is_gaming` survives, so the table supports a t-test and nothing else.	**0 to 2 points.** No join.
6. Missing values 6 points	**6 points.** Reports how many rows carry missing values and what you did with them.	**5 points.** Handled, but the count is not reported.	**3 to 4 points.** Rows with missing values dropped without saying how many. Dropping them silently is the error this step exists to catch.	**0 to 2 points.** Not handled.
7. Export 6 points	**6 points.** Saved as `twitch_analysis.RDS`.	**5 points.** Saved under another name, so the next two assignments cannot find it.	**3 to 4 points.** Saved as a CSV or another format that loses the column types.	**0 to 2 points.** Not exported.
8. Every step is in the script 6 points	**6 points.** Every change to the data happens in the script; nothing is done by hand in the console.	**5 points.** One step was done outside the script, but a comment says what and how.	**3 to 4 points.** Several steps were done outside the script.	**0 to 2 points.** The script does not rebuild the dataset.
Clean-session run 2 points	**2 points.** Runs start to finish after restarting R.	**1 point.** Runs after one small fix, such as a missing `library()` line.	**0 points.** Stops partway in a fresh session.	**0 points.** Runs only with objects already in your environment, or not at all.